

NOVA PROGRAM

Nova Functional Safety Requirements

Domain: Functional safety

Revision A | Synthetic extraction benchmark

2. Functional requirements

REQ-NVA-FSA-001 | Watchdog reaction

If the control task misses two consecutive execution windows, the independent watchdog shall reset the controller within 50 ms.

REQ-NVA-FSA-002 | Sensor plausibility

A disagreement greater than 8 deg C between redundant pack temperature channels shall set a diagnostic fault within 200 ms.

REQ-NVA-FSA-003 | Torque inhibition

While a charge connector is latched, commanded propulsion torque shall remain zero and vehicle displacement shall not exceed 150 mm.

3. Verification-oriented requirements

Identifier	Requirement title	Normative requirement
REQ-NVA-FSA-004	Safe-state persistence	After a crash latch is set, the controller shall hold both contactors open, suppress automatic restart, and preserve the latch across a 12 V power cycle.
REQ-NVA-FSA-005	Fault logging	Each ASIL-D safety reaction shall store the fault code, the first-failure timestamp with 10 ms resolution, and 2 seconds of pre-trigger sensor history.

Applicability notes

Context	Interpretation
Unless a requirement states otherwise	Limits apply to the complete production-intent assembly.
Use of shall	The complete sentence is normative; table columns do not split obligations.

4. Normative installation figure

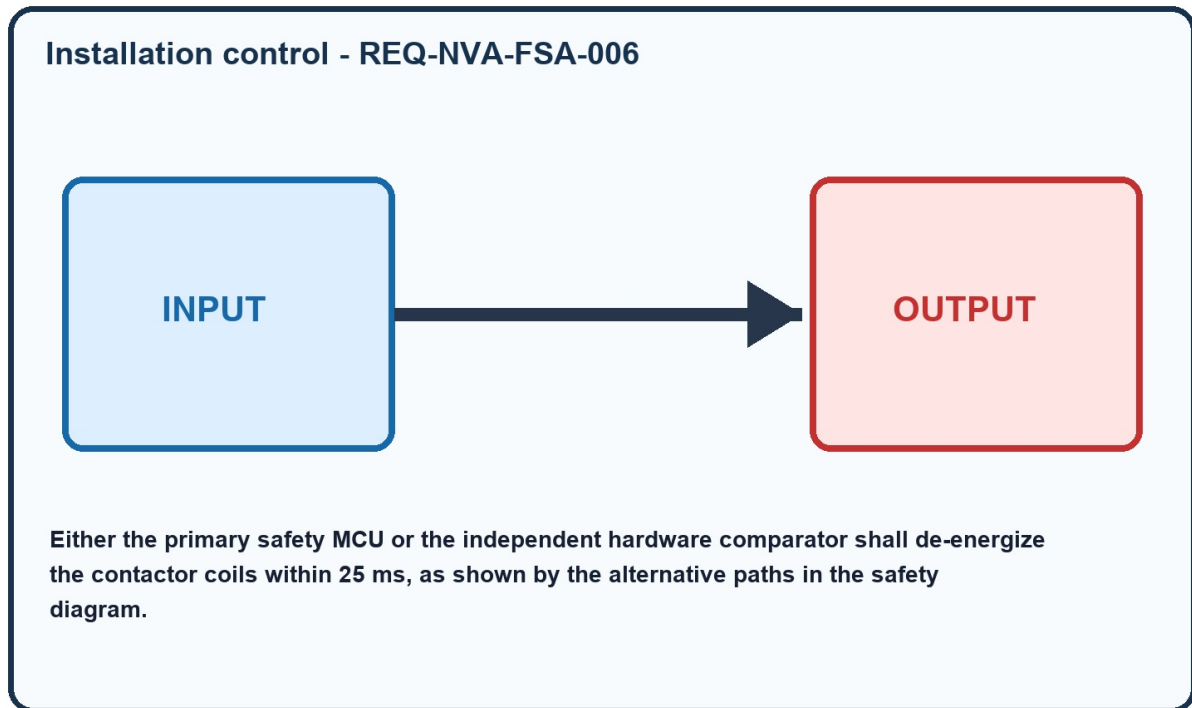


Figure 4-1. Colored geometry and embedded annotation are normative.

Engineering note

The figure annotation forms part of the controlled requirement baseline.

Color, direction, sequence, and geometric relationships must be interpreted visually.